Math 252
Spring 2018

Name ____________________

Test 1

1

a) Show how to calculate $C(12,4) = \binom{12}{4}$ **without a calculator. Show your work.**

b) Write a counting problem for which $\binom{12}{4}$ gives the anwer. (Keep your problem simple, do not use donuts.)

2 A sub shop is having a special: Three subs (one Italian, one veggie, and one meatball) for \$3. Alice can't resist and buys the three subs. Then it occurs to her that she is only hungry enough to eat one, so she decides to give away 2 of the 3 subs to some friends. Alice is meeting 10 friends to study. How many ways can she give two of the subs to two different of these friends (since they don't want more than one either)?

Note: Giving Bob the Italian sub and Claire the veggie sub is considered different from giving Claire the Italian sub and Bob the veggie sub.

3 Bitstrings

a) How many bitstrings of length 7 are there?

b) How many bitstrings of length 7 have *exactly* 4 1's?

c) How many bitstrings of length 7 have *at least* 4 1's?

d) How many bitstrings of length 7 have at least 4 *consecutive* 1's?

4 Below is an adjacency matrix representation of a graph G.

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}$$

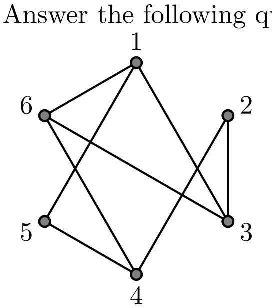
a) Is G directed or undirected? How can you tell?

b) How many vertices does G have? How can you tell?

c) How many edges does G have? How can you tell?

d) Does G have any self-loops? How can you tell?

5 Answer the following questions about the graph below.



a) What is an Euler circuit? Does this graph have an Euler circuit? How do you know?

b) Does the graph have an Euler path that is not a circuit? How do you know?

c) What is a bipartite graph? Is this graph bipartite? How do you know?

d) What is a planar graph? Is this graph planar? How do you know?

6 For **one** of K_5 or $K_{3,3}$ (your choice),

a) Draw the graph.

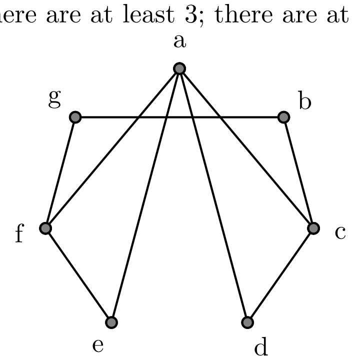
b) Explain how we know that the graph cannot be drawn in the plane with no edge crossings. Do not use Kuratowski's Theorem.

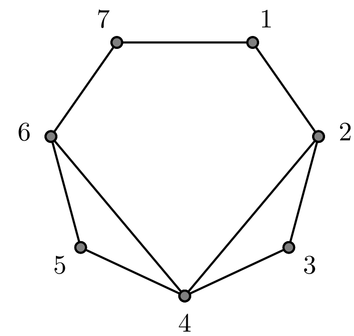
7 Draw a graph with 6 vertices, each of which has degree 3.

8 Explain why it is impossible to draw a graph with 7 vertices, all of which are degree 3.

9 How many isomorphisms are there between the following two graphs? **Explain.**

(If you do not know exactly how many there are, tell me as much as you know about how many there are. Examples: There are at least 3; there are at most 2, etc.)





Space for additional work.